The documentation and process conversion measures necessary to comply with this document shall be completed by 25 October 2006.

INCH-POUND

MIL-PRF-19500/225H
25 July 2006
SUPERSEDING
MIL-PRF-19500/225G
30 December 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON,
TYPES 2N1711, 2N1711S, 2N1890, AND 2N1890S, JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-5).

1.3 Maximum ratings unless otherwise specified $T_A$ = +25°C.

| Type (1) | $P_T$ (2) $T_C$ = +25°C | $P_T$-(2) $T_A$ = +25°C | $R_{\theta JA}$ (3) | $R_{\theta JC}$ (3) | $I_C$ | $V_{CBO}$ | $V_{CER}$ $R_{BE}$ = 10Ω | $V_{CEO}$ | $V_{EBO}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | °C/W | mA dc | V dc | V dc | V dc | V dc | °C |
| 2N1711 | 3.0 | 0.8 | 175 | 30 | 500 | 75 | 50 | 30 | 7 | -65 to +200 |
| 2N1890 | 3.0 | 0.8 | 175 | 30 | 500 | 100 | 80 | 60 | 7 | |

(1) Also applies to the corresponding "S" suffix device.
(2) For derating see figures 2 and 3.
(3) For thermal impedance curves see figures 4 and 5.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .



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* 1.4 Primary electrical characteristics.

| Limits | $h_{FE1}$ (1)/ | $h_{FE2}$ (1) | $\|h_{fe}\|$ | $V_{CE(SAT)}$ | | |
|---|---|---|---|---|---|---|
| | $V_{CE}$ = 10 V dc<br>$I_C$ = 10 µA dc | $V_{CE}$ = 10 V dc<br>$I_C$ = 150 mA dc | f = 20 MHz<br>$V_{CE}$ = 10 V dc<br>$I_c$ = 50 mA dc | 2N1711, S (1)<br>$I_C$ = 150 mA dc<br>$I_B$ = 15 mA dc | 2N1890, S (1)<br>$I_C$ = 150 mA dc<br>$I_B$ = 15 mA dc | 2N1890, S (1)<br>$I_C$ = 50 mA dc<br>$I_B$ = 5.0 mA dc |
| | | | | V dc | V dc | V dc |
| Min | 20 | 100 | 3.5 | | | |
| Max | | 300 | 12 | 1.5 | 5.0 | 1.2 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

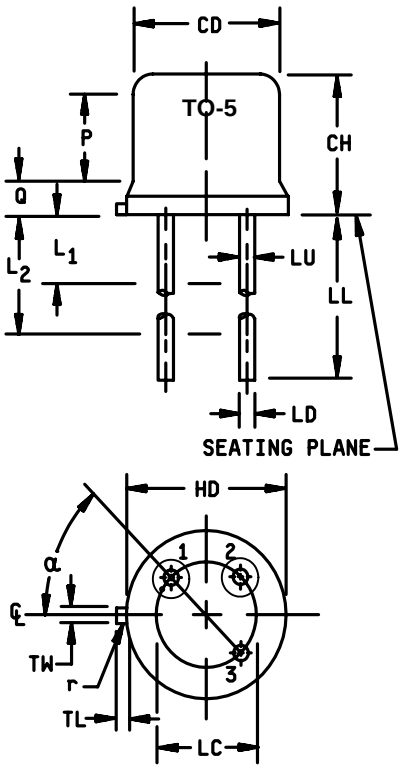
(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Ltr | Dimensions | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TYP | | 5.08 TYP | | 7 |
| LD | .016 | .021 | 0.41 | 0.53 | 6 |
| LL | See notes 7, 9, and 10 | | | | |
| LU | .016 | .019 | 0.41 | 0.48 | 7 |
| $L_1$ | | .050 | | 1.27 | 7 |
| $L_2$ | .250 | | 6.35 | | 7 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | |
| r | | .010 | | 0.254 | 8 |
| TL | .029 | .045 | 0.74 | 1.14 | 4 |
| TW | .028 | .034 | 0.71 | 0.86 | 3 |
| $\alpha$ | 45° TP | | 45° TP | | 6 |
| Term 1 | Emitter | | | | |
| Term 2 | Base | | | | |
| Term 3 | Collector | | | | |




NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r maximum, TW must be held to a minimum length of .021 inch (0.53 mm).
4. TL measured from maximum HD.
5. CD shall not vary more than ±.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 - .055 inch (1.37 - 1.40 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by gauge and gauging procedure.
7. LU applies between $L_1$ and $L_2$. LD applies between $L_2$ and L minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. r (radius) applies to both inside corners of tab.
9. For transistor types 2N1711S and 2N1890S, LL is .500 inch (12.70 mm) minimum, and .750 inch (19.50 mm) maximum (TO-5).
10. For transistor types 2N1711 and 2N1890, LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum (TO-5).
11. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions (TO-5).



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-5).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (list applicable JANTX level). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement<br><br>JANTX level |
|---|---|
| (1) 3c | Thermal impedance (see 4.3.2), method 3131 of MIL-STD-750. |
| 9 | Not applicable |
| 11 | $h_{FE2}$; $I_{CBO1}$ |
| 12 | See 4.3.1 |
| 13 | $\Delta I_{CBO1}$ = 100 percent of initial value or 5 nA dc; whichever is greater;<br>$\Delta h_{FE2}$ = ±15 percent of initial value; |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Burn-in conditions. Burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum, $T_A$ ambient rated as defined in 1.3. NOTE: No heat sink or forced air cooling of the devices shall be permitted.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The test conditions for $Z_{\theta JX}$ shall be those used in 4.3.2.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified in 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.



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4.4.2.1 Group B inspection, JAN and JANTX.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: Test condition B, 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is a minimum of 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

* 4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN and JANTX samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test ( group B for JAN and JANTX) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Lead fatigue: Test condition E. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2). |
| C6 | 1026 | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal 5/<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements | | Table I, subgroup 2 | | | | |
| Bond strength 3/ | 2037 | Precondition<br>$T_A$ = +250°C at t = 24 hours or<br>$T_A$ = +300°C at t = 2 hours<br>n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage, collector to base | 3001 | Bias condition D;<br>$I_C$ = 100 μA dc; | $V_{(BR)CBO}$ | | | |
| 2N1711, 2N1711S | | | | 75 | | V dc |
| 2N1890, 2N1890S | | | | 100 | | V dc |
| Breakdown voltage, emitter to base | 3026 | Bias condition D;<br>$I_E$ = 100 μA dc; | $V_{(BR)EBO}$ | 7 | | V dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 30 mA dc; pulsed (see 4.5.1). | $V_{(BR)CEO}$ | | | |
| 2N1711, 2N1711S | | | | 30 | | V dc |
| 2N1890, 2N1890S | | | | 60 | | V dc |
| Breakdown voltage, collector to emiiter | 3011 | Bias condition D; $I_C$ = 100 mA dc; pulsed (see 4.5.1). $R_{BE}$ = 10Ω. | $V_{(BR)CER}$ | | | |
| 2N1711, 2N1711S | | | | 50 | | V dc |
| 2N1890, 2N1890S | | | | 80 | | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 – Continued. | | | | | | |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO1}$ | | | |
| 2N1711, 2N1711S | | $V_{CB}$ = 60V dc | | | 10 | nA dc |
| 2N1890, 2N1890S | | $V_{CB}$ = 80 V dc | | | 10 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO}$ | | 5.0 | nA dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | | |
| 2N1711, 2N1711S | | | | | 1.5 | V dc |
| 2N1890, 2N1890S | | | | | 5.0 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 50 mA dc; $I_B$ = 5 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | | V dc |
| 2N1890, 2N1890S | | | | | 1.2 | V dc |
| Base emitter saturation voltage | 3066 | Test condition A, $I_C$ = 150 mA dc, $I_B$ = 15 mA dc, pulsed (see 4.5.1). | $V_{BE(sat)1}$ | | 1.3 | V dc |
| Base emitter saturation voltage | 3066 | Test condition A, $I_C$ = 50 mA dc, $I_B$ = 5 mA dc, pulsed (see 4.5.1). | $V_{BE(sat)2}$ | | | V dc |
| 2N1890, 2N1890S | | | | | 0.9 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 µA dc pulsed (see 4.5.1). | $h_{FE1}$ | 20 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc pulsed (see 4.5.1). | $h_{FE2}$ | 100 | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc pulsed (see 4.5.1). | $h_{FE3}$ | | | |
| 2N1711, 2N1711S | | | | 50 | | |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO2}$ | | | |
| 2N1711, 2N1711S | | $V_{CB}$ = 60 V dc | | | 10 | µA dc |
| 2N1890, 2N1890S | | $V_{CB}$ = 80 V dc | | | 15 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc, pulsed (see 4.5.1). | $h_{FE4}$ | 35 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward current transfer ratio | 3206 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc<br>f = 1KHz | $h_{fe}$ | 80 | 200 | |
| | | $V_{CE}$ = 10 V dc; $I_C$ = 5 mA dc<br>f = 1KHz. | $h_{fe}$ | 90 | 270 | |
| Magnitude of small-signal short-circuit forward current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 50 mA dc;<br>f = 20 MHz | $\lvert h_{fe} \rvert$ | 3.5 | 12 | |
| Small-signal short- circuit input impedance | 3201 | $V_{CB}$ = 10 V dc;<br>$I_C$ = 5 mA dc | $h_{ib}$ | 4 | 8 | ohms |
| Small-signal short-circuit input Admittance | 3216 | $V_{CB}$ = 10 V dc;<br>$I_C$ = 5 mA dc | $h_{ob}$ | | | |
| 2N1711, 2N1711S<br>2N1890, 2N1890S | | | | 0.0<br>0.0 | 1.0<br>.3 | µmhos<br>µmhos |
| Small-signal open-circuit reverse voltage transfer ratio | 3211 | $V_{CB}$ = 10 V dc;<br>$I_C$ = 5 mA dc | $h_{rb}$ | | | |
| 2N1711, 2N1711S<br>2N1890, 2N1890S | | | | | $5x10^{-4}$<br>$1.5x10^{-4}$ | µmhos<br>µmhos |
| Open circuit output capacitance | | $V_{CB}$ = 10 V dc; $I_E$ = 0 mA dc<br>100kHz ≤ f ≤ 1 mHz | $C_{obo}$ | | | |
| 2N1711, 2N1711S<br>2N1890, 2N1890S | | | | 8<br>5 | 25<br>15 | pF<br>pF |
| Pulse response | 3251 | Test condition A, except test circuit and pulse requirements.<br>See figure 6 herein. | $t_{on} + t_{off}$ | | 30 | ns |
| Subgroup 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.



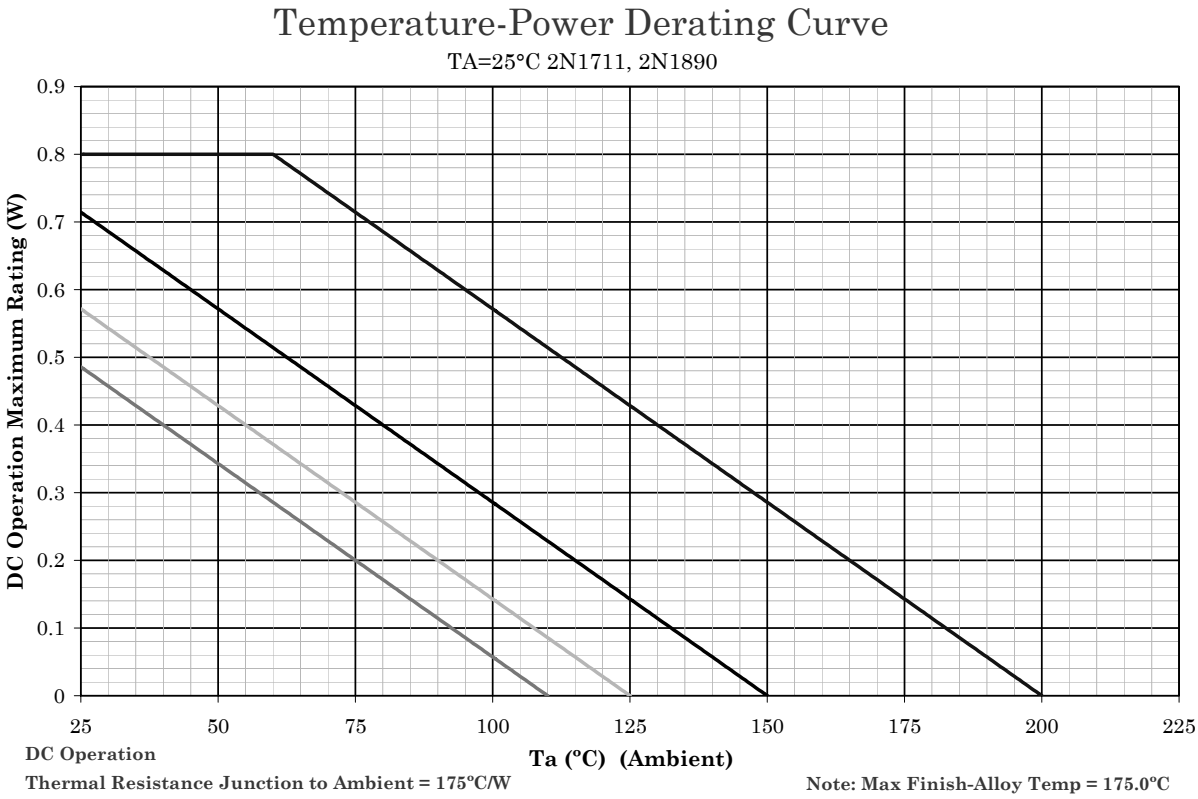
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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles<br>Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| Electrostatic discharge (ESD) | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |



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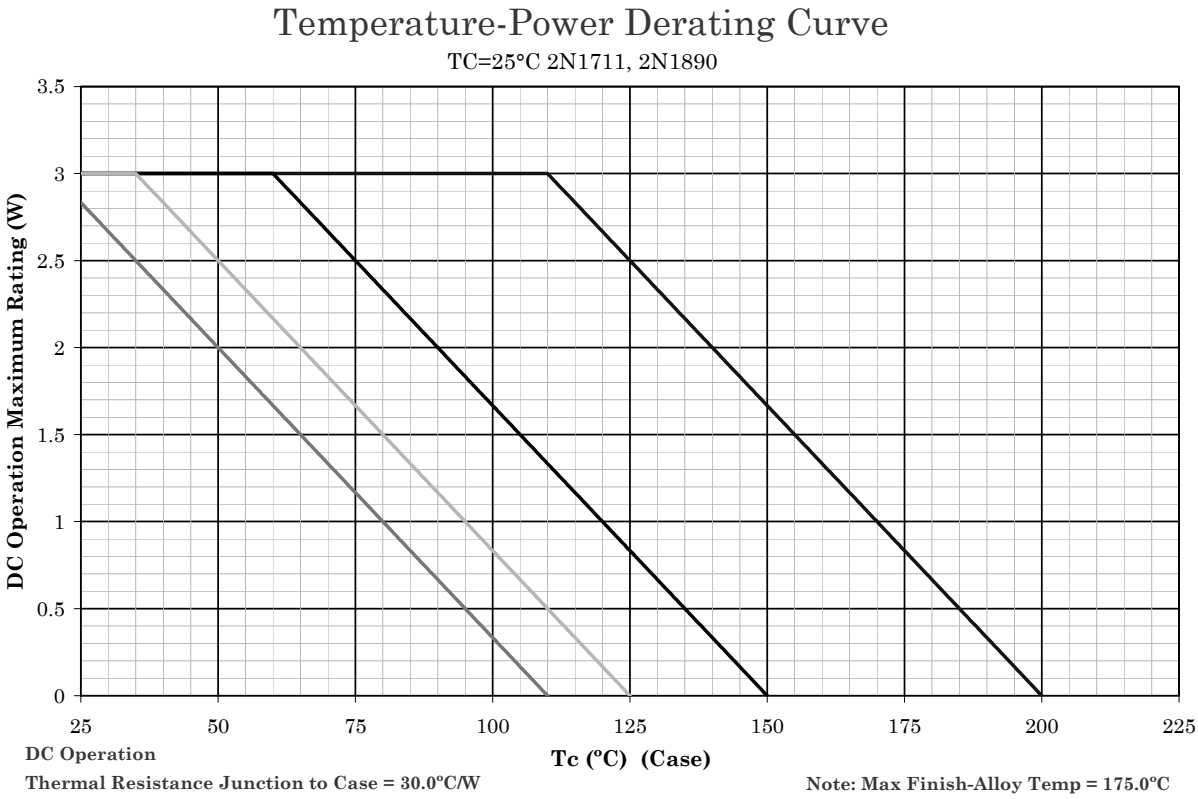

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 2. Temperature-power derating for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).



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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 3. Temperature-power derating for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).



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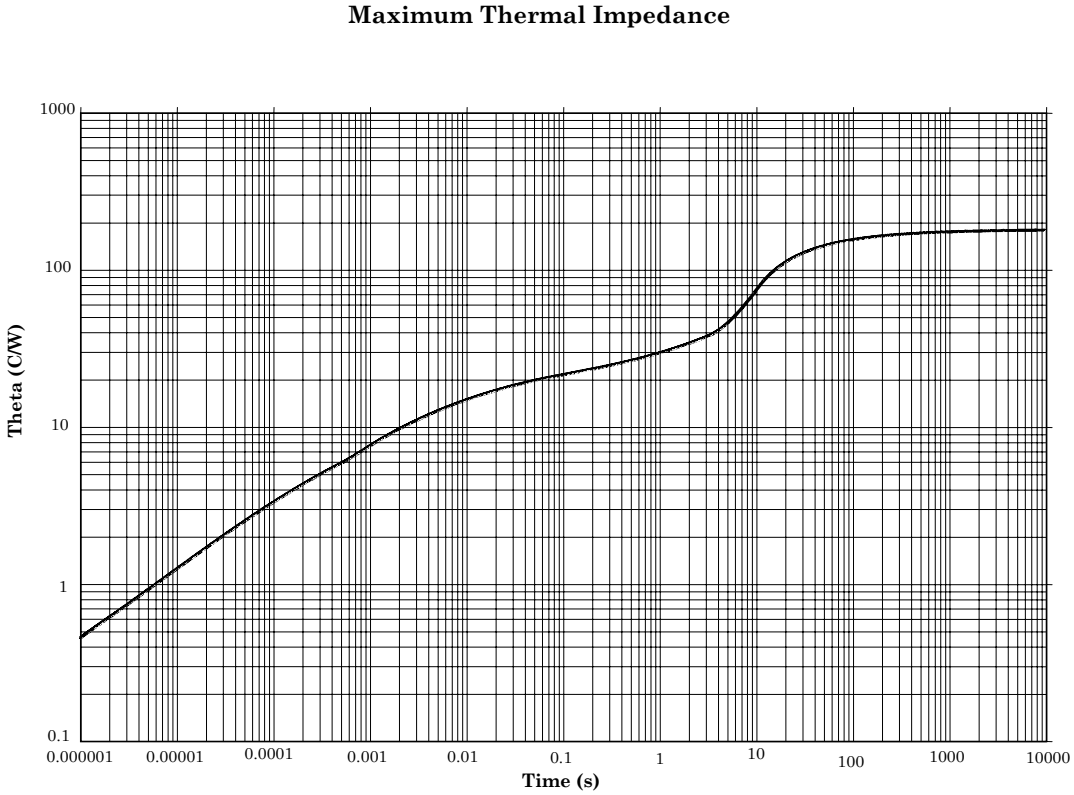



FIGURE 4. Thermal impedance graph ($R_{\theta JA}$) for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).



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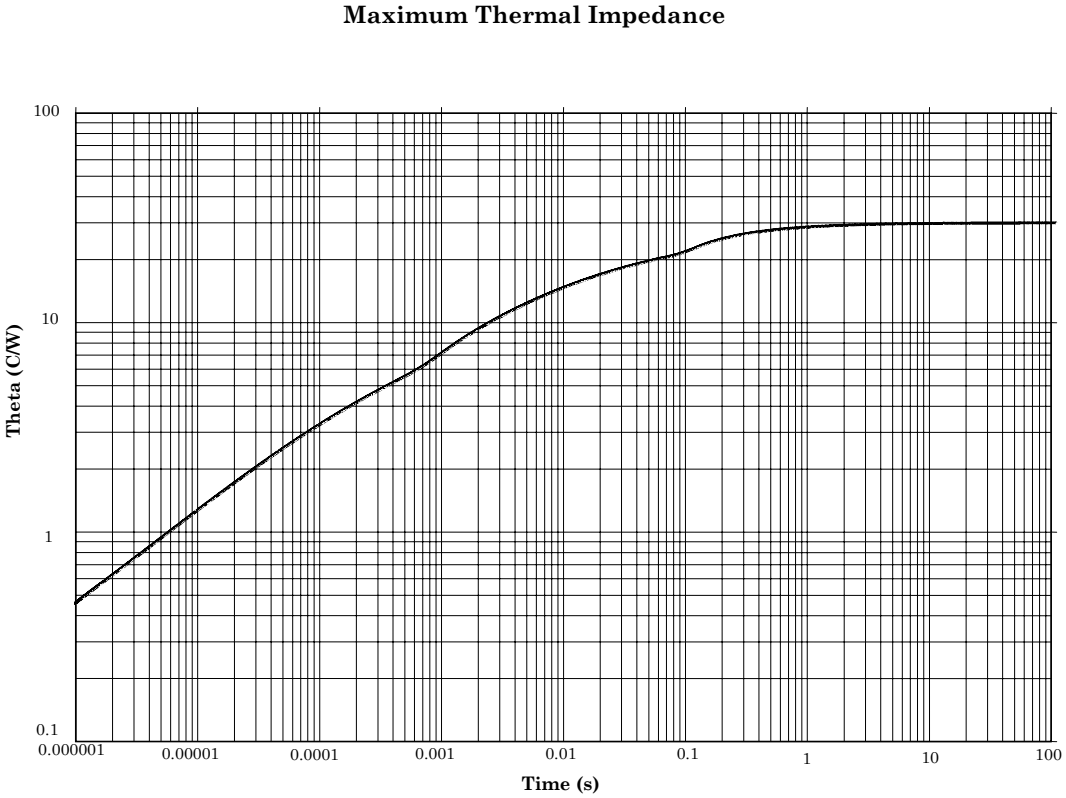
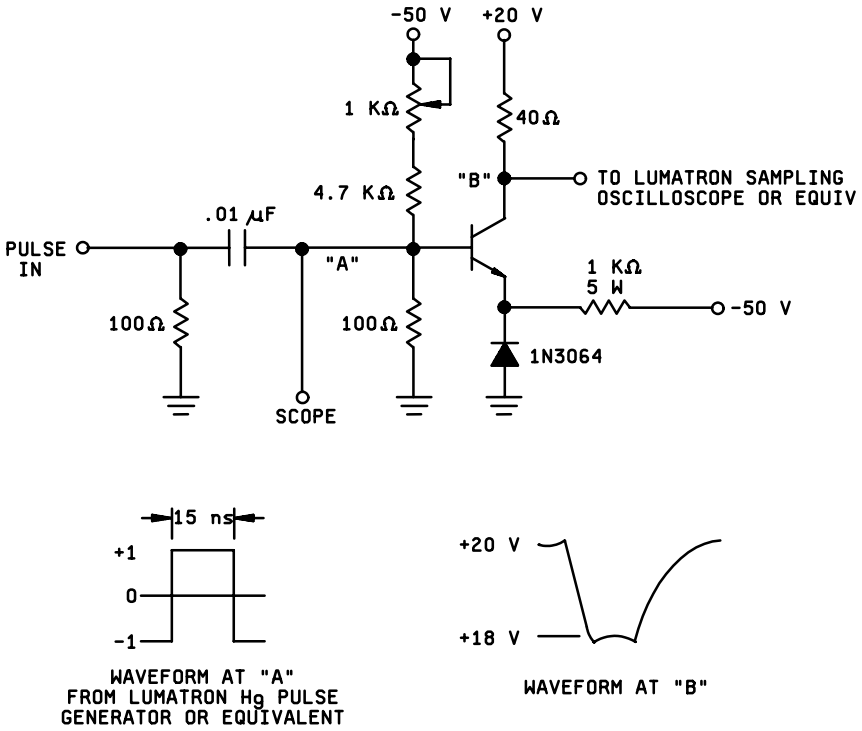
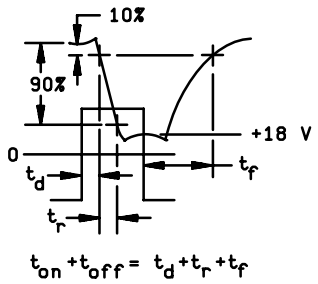



FIGURE 5. Thermal impedance graph ($R_{\theta JC}$) for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).



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NOTES:

1. The rise time ($T_r$) of the applied pulse shall be 2.0 ≤ ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: $Z_{in} \geq 100$ kΩ, $C_{in} \leq 12$ pF, rise time ≤ 2 ns.

FIGURE 6. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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